SECR2043 OPERATING SYSTEMS

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Section : 03

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	**Output**
	pwd	/home/lam
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	**Output**
	ls -a	. .. .bash_logout .bashrc .profile example.txt share
3.	Create a new directory named "os_lab" in your home directory.	
	Command	**Output**
	mkdir os_lab	No Output
4.	Change your current working directory to "os_lab".	
	Command	**Output**
	cd os_lab	Directory Changes: lam@secr2043:~/os_lab$
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	**Output**
	touch hello.txt echo "Hello, world!" > hello.txt	No Output

6.	Display the content of "hello.txt".	
	Command	**Output**
	cat hello.txt	Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	**Output**
	cp hello.txt hello_copy.txt	No Output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	**Output**
	mv hello_copy.txt hello_bak.txt	No Output
9.	Delete "hello.txt".	
	Command	**Output**
	rm hello.txt	No Output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	**Output**
	~ ls -lR ~	/home/lam: total 8 -rw-r--r-- 1 lam lam 0 Apr 21 09:10 example.txt drwxr-xr-x 2 lam lam 4096 Apr 21 09:25 os_lab drwxr-xr-x 2 lam lam 4096 Apr 21 09:10 share /home/lam/os_lab: total 4 -rw-r--r-- 1 lam lam 14 Apr 21 09:22 hello_bak.txt /home/lam/share: total 0 -rw-r--r-- 1 lam lam 0 Apr 21 09:10 textfile01.txt
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

		Command	Output
1.	Create an empty file named "hello.txt" in the current working directory.	touch hello.txt	No Output
2.	Write the text "Hello World" to "hello.txt" using echo.	echo "Hello, world" > hello.txt	No Output
3.	Append the text "This is a test" to "hello.txt" using echo.	echo "This is a test" >> hello.txt	No Output
4.	Display the contents of "hello.txt" using cat.	cat hello.txt	Hello World This is a test
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	echo "Goodbye World" > bye.txt	No Output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	cat hello.txt bye.txt	Hello World This is a test Goodbye World
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	cat hello.txt bye.txt > greeting.txt	No Output

<table>
<tr><td rowspan="3">8.</td><td colspan="2">Display the contents of "greeting.txt" using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat greeting.txt</td><td>Hello World
This is a test
Goodbye World</td></tr>
<tr><td rowspan="3">9.</td><td colspan="2">Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>nano greeting.txt</td><td>GNU nano 7.2 is opened</td></tr>
<tr><td rowspan="3">10.</td><td colspan="2">Display the modified contents of "greeting.txt" using cat.</td></tr>
<tr><td>Command</td><td>Output</td></tr>
<tr><td>cat greeting.txt</td><td>Hello Lam
This is a test
Goodbye Lam</td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
</table>

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc.
For each question, write down the command and the output (if any) in the answer sheet.

1. Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".

Command	Output
nano hello.c	No Output

Code:
```
#include <stdio.h>

int main() {
        printf("Hello, World!\n");
        return 0;
}
```

2. Compile the program "hello.c" using gcc and generate an executable file named "hello".

Command	Output
gcc hello.c -o hello	No Output

3. Run the executable file "hello" and display its output.

Command	Output
./hello	Hello, World!

4. Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".

Command	Output
nano sum.c	No Output

Code:
```
#include <stdlib.h>
#include <stdio.h>

int main(int argc, char *argv[]){
        int x = atoi(argv[1]);
        int y = atoi(argv[2]);
        int sum = x + y;
        printf("Sum: %d\n", sum);
}
```

5. Compile the program "sum.c" using gcc and generate an executable file named "sum".

Command	Output
gcc sum.c -o sum	No Output

6. Run the executable file "sum" with 10 and 20 as arguments and display its output.

Command	Output
./sum 10 20	Sum: 30

7. Run the executable file "sum" with -5 and 15 as arguments and display its output.

Command	Output
./sum -5 15	Sum: 10

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	

Code:

```
#include <stdio.h>

int main() {
	char string[100];

	printf("Enter a string: ");
	fgets(string, 100, stdin);
	printf("%s", string);
}
```

	Command	Output
	nano echo.c	No Output

9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	**Output**
	gcc echo.c -o echo	No Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	**Output**
	./echo	Enter a string: Hello World Hello World
	Total Mark:	